

Intermediate Level Statistics

Theoretical Foundations for Statistics and Machine Learning

1. Poisson Distribution as a Limiting Case of Binomial Distribution

Idea

The **Poisson distribution** can be obtained from the **Binomial distribution** when the number of trials becomes very large and the probability of success becomes very small.

Conditions:

$$\begin{aligned}n &\rightarrow \infty \\p &\rightarrow 0 \\np &= \lambda \quad (\text{constant})\end{aligned}$$

Where:

- n = number of trials
- p = probability of success
- λ = average number of occurrences

Under these conditions:

Binomial Distribution $\rightarrow$ Poisson Distribution

Result (MGF Form)

The moment generating function becomes:

$$M_X(t) = e^{\lambda(e^t - 1)}$$

This is the **MGF of Poisson distribution**.

Interpretation

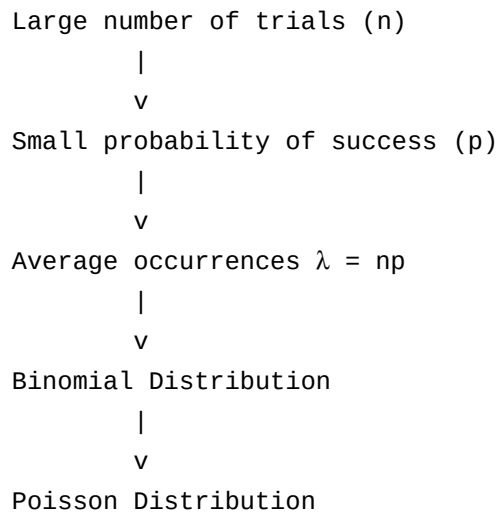
Poisson distribution is used when:

- events are **rare**
- number of trials is **very large**
- probability of each event is **very small**

Examples:

- number of defects in manufactured products
- number of accidents per day
- number of emails received per hour

Concept Diagram



Example

Suppose a factory produces **10,000 components** per day. Probability that a component is defective:

$$p = 0.0002$$

Expected number of defective components:

$$\lambda = np$$

$$\lambda = 10000 \times 0.0002$$

$$\lambda = 2$$

Thus the **number of defective components follows Poisson distribution with $\lambda = 2$.**

Sample Problem 1

A factory produces 2000 items daily. The probability that an item is defective is 0.001. Find the average number of defective items.

Solution

Given:

$$n = 2000$$

$$p = 0.001$$

Compute:

$$\lambda = np$$

$$\lambda = 2000 \times 0.001$$

$$\lambda = 2$$

Therefore the **Poisson distribution parameter is $\lambda = 2$.**

2. Central Limit Theorem (CLT)

Definition

*The distribution of the sum or average of a large number of independent random variables approaches a **normal distribution**, regardless of the original distribution.*

This theorem is one of the most important results in probability and statistics.

Mathematical Form

If $X_1, X_2, X_3, \dots, X_n$ are independent random variables with:

$$\text{Mean} = \mu$$

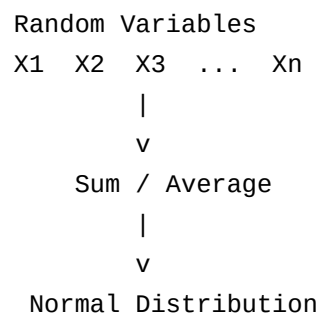
$$\text{Variance} = \sigma^2$$

then the standardized variable:

$$Z = (X - \mu) / \sigma$$

follows approximately a **standard normal distribution** when n is large.

Concept Diagram



3. Applications of Central Limit Theorem

The CLT is useful because it:

1. Helps compute probabilities of sums of random variables
2. Explains why many natural phenomena follow a **normal distribution**
3. Allows approximation of complicated distributions using **normal distribution**

Practical Examples

CLT is used in:

- Machine learning data analysis
- Statistical sampling
- Quality control in manufacturing

- Population studies
 - Financial modeling
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4. Normal Approximation of Binomial Distribution

When the number of trials is large, the **Binomial distribution can be approximated by a Normal distribution.**

Conditions:

- n is large
- p is not extremely small

Mean:

$$\mu = np$$

Variance:

$$\sigma^2 = npq$$

Standard deviation:

$$\sigma = \sqrt{(npq)}$$

Example

A coin is tossed **300 times**:

$$p = 0.5$$

$$n = 300$$

$$q = 0.5$$

Mean:

$$\mu = np$$

$$\mu = 300 \times 0.5$$

$$\mu = 150$$

Variance:

$$\sigma^2 = npq$$

$$\sigma^2 = 300 \times 0.5 \times 0.5$$

$$\sigma^2 = 75$$

Standard deviation:

$$\sigma = \sqrt{75}$$

$$\sigma \approx 8.66$$

Thus the **binomial distribution can be approximated by a normal distribution** with Mean = 150 and Variance = 75.

Standardization Formula

To convert to standard normal variable:

$$Z = (X - \mu) / \sigma$$

Where:

X = observed value

μ = mean

σ = standard deviation

Sample Problem 2

A coin is tossed **300 times**. Find the probability that the number of heads lies between **140 and 150**.

Step 1: Find Mean

$$\mu = np$$

$$\mu = 300 \times 0.5$$

$$\mu = 150$$

Step 2: Find Variance

$$\sigma^2 = npq$$

$$\sigma^2 = 300 \times 0.5 \times 0.5$$

$$\sigma^2 = 75$$

Standard deviation:

$$\sigma = \sqrt{75}$$

$$\sigma \approx 8.66$$

Step 3: Convert to Z values

For $X = 140$:

$$Z = (140 - 150) / \sqrt{75} \approx -1.15$$

For $X = 150$:

$$Z = 0$$

Step 4: Use Normal Table

$$P(-1.15 < Z < 0) = P(0 < Z < 1.15)$$

From standard normal table:

$$P = 0.3749$$

Intermediate Level Summary

Topics Covered

- ✓ Poisson distribution as limiting case of binomial distribution
- ✓ Conditions for Poisson approximation
- ✓ Central Limit Theorem (CLT)
- ✓ Applications of CLT
- ✓ Normal approximation of binomial distribution